

# **SOCIO-ECONOMIC STUDY OF THE FISHERMEN COMMUNITY AT COLACHEL FISHING HARBOUR**

**A PROJECT**

*Submitted by*

**JOEL RAJ R**

**Reg.No.: 24084016513411108**

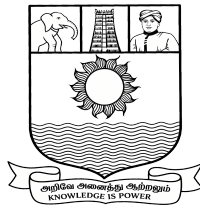
*in partial fulfillment for the award of the degree of*

**MASTER OF SCIENCE IN  
STATISTICS**



**DEPARTMENT OF STATISTICS  
MANONMANIAM SUNDARANAR UNIVERSITY  
TIRUNELVELI - 627 012**

**APRIL 2026**



**MANONMANIAM SUNDARANAR UNIVERSITY**  
**TIRUNELVELI - 627 012**

---

**BONAFIDE CERTIFICATE**

Certified that this project titled "**SOCIO-ECONOMIC STUDY OF THE FISHERMEN COMMUNITY AT COLACHEL FISHING HARBOUR**" is the bonafide work of **JOEL RAJ R (Reg.No.: 24084016513411108)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

**Signature of the HOD**  
with date

**Signature of the Supervisor**  
with date

# **SOCIO-ECONOMIC STUDY OF THE FISHERMEN COMMUNITY AT COLACHEL FISHING HARBOUR**

**JOEL RAJ R**

## **Abstract**

This study examines the socio-economic conditions of 114 male fishermen in and around Colachel Fishing Harbor in Kanyakumari district, Tamil Nadu. The study focuses on demographic profile, income variation, expenditure pattern, fishing safety perceptions, and opinions on government welfare measures. The findings reveal that most fishermen have low educational attainment and long years of fishing experience. Income is found to vary based on factors such as fishing experience and occupational role. Loan repayment constitutes the major share of household expenditure, indicating significant financial burden. The study also shows that there is no notable difference in safety perceptions across different fishing roles. However, a majority of respondents expressed dissatisfaction with government welfare measures related to deep-sea safety. Overall, the study highlights the economic vulnerability of the fishing community and emphasizes the need for effective policy implementation and improved support systems.

**Keywords:** Socio-Economic Status, Colachel Fishing Harbor, ANOVA, Regression, Expenditure Pattern, Government Welfare

## **ACKNOWLEDGMENT**

I express my sincere gratitude to the Almighty God for His grace and blessings, which enabled me to successfully complete this project.

I am deeply indebted to my guide, **Dr. K. Manoj**, Assistant Professor, Department of Statistics, Manonmaniam Sundaranar University, for his invaluable guidance, constant encouragement, and insightful suggestions throughout the course of this study. His support has been instrumental in shaping this work.

I extend my heartfelt thanks to **Dr. P. Arumugam**, Professor and Head, and all the faculty members of the Department of Statistics, especially **Dr. R. Sasikumar**, Assistant Professor, and **Dr. V. Deneshkumar**, Assistant Professor, for their continuous support, encouragement, and academic guidance during my postgraduate study.

I gratefully acknowledge the support and encouragement of my parents, friends, and well-wishers, whose motivation and assistance helped me complete this project successfully.

Finally, I thank all those who have directly or indirectly contributed to the successful completion of this work.

**JOEL RAJ R**

## TABLE OF CONTENTS

|                                                            |             |
|------------------------------------------------------------|-------------|
| <b>ABSTRACT</b>                                            | <b>ii</b>   |
| <b>ACKNOWLEDGMENT</b>                                      | <b>iii</b>  |
| <b>LIST OF TABLES</b>                                      | <b>vii</b>  |
| <b>LIST OF FIGURES</b>                                     | <b>viii</b> |
| <b>1 INTRODUCTION</b>                                      | <b>1</b>    |
| 1.1 Introduction . . . . .                                 | 1           |
| 1.2 The Fishing Community: An Overview . . . . .           | 2           |
| 1.3 Socio-Economic Vulnerabilities . . . . .               | 3           |
| 1.4 Role of Government Welfare Schemes . . . . .           | 4           |
| 1.5 Significance of the Study . . . . .                    | 5           |
| 1.6 Limitations of the Study . . . . .                     | 5           |
| 1.7 Objectives of the Study . . . . .                      | 6           |
| <b>2 REVIEW OF LITERATURE</b>                              | <b>7</b>    |
| 2.1 Socio-Economic Profile of Fishermen in India . . . . . | 7           |
| 2.2 Education and Income in Fishing Communities . . . . .  | 8           |
| 2.3 Expenditure Patterns and Indebtedness . . . . .        | 9           |
| 2.4 Safety, Governance, and Government Schemes . . . . .   | 10          |
| 2.5 Fishing Safety and Technology Adoption . . . . .       | 11          |
| <b>3 DATA DESCRIPTION AND METHODOLOGY</b>                  | <b>13</b>   |
| 3.1 Study Area . . . . .                                   | 13          |
| 3.2 Study Design . . . . .                                 | 14          |

---

|          |                                                              |           |
|----------|--------------------------------------------------------------|-----------|
| 3.3      | Sampling Technique . . . . .                                 | 14        |
| 3.4      | Data Collection . . . . .                                    | 15        |
| 3.5      | Data Description . . . . .                                   | 16        |
| 3.6      | Ethical Considerations . . . . .                             | 16        |
| 3.7      | Statistical Methods . . . . .                                | 17        |
| 3.7.1    | Descriptive Statistics . . . . .                             | 17        |
| 3.7.2    | One-Way Analysis of Variance (ANOVA) . . . . .               | 17        |
| 3.7.3    | Multiple Linear Regression Analysis . . . . .                | 18        |
| 3.7.4    | Chi-Square Test . . . . .                                    | 19        |
| 3.7.5    | Frequency . . . . .                                          | 20        |
| 3.8      | Tools for Data Analysis . . . . .                            | 20        |
| <b>4</b> | <b>DATA ANALYSIS AND INTERPRETATION</b>                      | <b>21</b> |
| 4.1      | Demographic Profile of the Fishermen . . . . .               | 21        |
| 4.1.1    | Gender and Marital Status . . . . .                          | 21        |
| 4.1.2    | Educational Qualification . . . . .                          | 22        |
| 4.1.3    | Age, Family Size, and Experience . . . . .                   | 23        |
| 4.2      | One-Way ANOVA . . . . .                                      | 24        |
| 4.2.1    | Annual Income by Educational Qualification . . . . .         | 24        |
| 4.2.2    | Annual Income by Fishing Mode . . . . .                      | 26        |
| 4.2.3    | Annual Income by Role in Fishing Activity . . . . .          | 27        |
| 4.3      | Multiple Regression Analysis of Annual Income . . . . .      | 28        |
| 4.4      | Household Expenditure Pattern . . . . .                      | 31        |
| 4.4.1    | Savings and Income Stability . . . . .                       | 32        |
| 4.5      | Safety Preceptions by Fishing Role . . . . .                 | 33        |
| 4.5.1    | New Techniques and Climate Information Readiness . . . . .   | 34        |
| 4.6      | Government Welfare and Deep-Sea Safety Perceptions . . . . . | 35        |
| 4.6.1    | Access to Government Welfare Schemes . . . . .               | 35        |
| 4.6.2    | Perception of Government Steps for Deep-Sea Safety . . . . . | 36        |
| 4.6.3    | Suggested Government Measures . . . . .                      | 37        |
| 4.7      | Self-Reported Economic Status . . . . .                      | 38        |

---

|                                      |               |
|--------------------------------------|---------------|
| <b>5 SUMMARY AND CONCLUSION</b>      | <b>39</b>     |
| 5.1 Summary of Results . . . . .     | 39            |
| 5.2 Conclusion . . . . .             | 41            |
| 5.3 Policy Recommendations . . . . . | 41            |
| REFERENCES . . . . .                 | 42            |
| <br><b>APPENDIX</b>                  | <br><b>45</b> |

## **List of Tables**

|      |                                                          |    |
|------|----------------------------------------------------------|----|
| 4.1  | Gender Distribution of Respondents . . . . .             | 22 |
| 4.2  | Marital Status of Respondents . . . . .                  | 22 |
| 4.3  | Annual Income by Role in Fishing Activity . . . . .      | 23 |
| 4.4  | Descriptive statistics of Demographic Profile . . . . .  | 24 |
| 4.5  | Education Level and Mean Income of Respondents . . . . . | 25 |
| 4.6  | Income Distribution by Fishing Mode . . . . .            | 26 |
| 4.7  | Annual income by fishing activity . . . . .              | 27 |
| 4.8  | ANOVA Results for Regression Model . . . . .             | 28 |
| 4.9  | Regression Coefficients . . . . .                        | 28 |
| 4.10 | Expenditure Distribution of Respondents . . . . .        | 31 |
| 4.11 | Income Stability Analysis . . . . .                      | 33 |
| 4.12 | Opinion by Fishing Role . . . . .                        | 34 |
| 4.13 | Adoption and Awareness of Fishing Practices . . . . .    | 35 |
| 4.14 | Welfare Receipt Status . . . . .                         | 36 |
| 4.15 | Steps taken for welfare . . . . .                        | 36 |
| 4.16 | Economic Status Distribution . . . . .                   | 38 |



## **List of Figures**

|                                                                                                       |    |
|-------------------------------------------------------------------------------------------------------|----|
| 4.1 Histogram of standardized residuals showing approximate normal distribution . . . . .             | 29 |
| 4.2 Normal P-P plot of standardized residuals indicating normality assumption . . . . .               | 29 |
| 4.3 Scatter plot of standardized residuals versus predicted values showing homoscedasticity . . . . . | 30 |
| 4.4 Household Income Expenditure Pattern . . . . .                                                    | 32 |
| 4.5 Fishermen's Perception of Government Support for Deep-Sea Safety . . . . .                        | 37 |

## **CHAPTER - 1**

### **INTRODUCTION**

*"The sea is not a bargain basement." Jacques Yves Cousteau*

This chapter provides a broad overview of the socio-economic conditions of the fishermen community in India, with a particular focus on the context of coastal Tamil Nadu. It covers the nature of fishing as a livelihood, the economic and social vulnerabilities faced by fishing households, the role of government welfare schemes, and the significance of studying this community. A understanding of these dimensions is essential for interpreting the findings presented in subsequent chapters.

#### **1.1 Introduction**

Fishing is one of the oldest and most vital occupations in human civilization, especially in coastal nations like India. The fisheries sector in India is a significant contributor to the national economy, providing livelihoods to approximately 28 million people engaged in fishing and allied activities, and contributing around 1.24% to the national Gross Domestic Product (GDP). Coastal fishermen, who constitute a large segment of this workforce, live in close proximity to the sea, and their daily lives are shaped by the rhythm of the ocean, seasonal fishing cycles, and the socio-economic structures of their communities.

Despite their critical economic role, fishermen communities in India particularly in Tamil Nadu remain among the most economically vulnerable sections of the population. The fishing community around Colachel Fishing

Harbor in Kanyakumari District is representative of many such communities along the Indian coastline. These fishermen encounter a complex matrix of economic precarity, limited educational attainment, seasonal income fluctuations, indebtedness, and insufficient access to formal financial systems and government benefits.

Colachel, located at the southernmost tip of India, is one of the major fishing harbors in Tamil Nadu. It is home to thousands of fishing families who rely primarily on marine fishing as their main source of income. The harbor facilitates both artisanal and mechanized fishing activities, ranging from traditional country boats to large trawlers and mechanized vessels. The community is deeply rooted in its cultural and occupational heritage, with fishing experience often being passed down from generation to generation.

The present study was motivated by the need to understand the socio-economic realities of this community in a systematic and data-driven manner. While several qualitative accounts and policy documents have highlighted the challenges faced by coastal fishermen, rigorous quantitative studies examining the linkages between education, income, expenditure patterns, and perceptions of government support remain limited for this specific region. This study aims to fill that gap by providing empirical evidence that can inform policy decisions and community development interventions.

## **1.2 The Fishing Community: An Overview**

The fishing community in India is heterogeneous, comprising a wide variety of occupational roles, income levels, and social structures. In the context of Colachel and similar harbors, fishing activities can be broadly classified according to the mode of fishing (traditional versus mechanized), the role of the individual in the fishing enterprise (worker, shareholder, or owner), and the type of income arrangement (daily wages, weekly payments, or monthly salary).

Traditional fishing methods typically involve the use of country boats and non-motorized equipment, whereas mechanized fishing employs trawlers and motorized vessels equipped with modern gear. The shift towards mechanization has brought significant economic changes, including higher income potential but also greater financial risk, investment requirements, and operational costs such as fuel, net maintenance, and boat upkeep.

Ownership structures within the fishing community also vary. While some fishermen own their boats and equipment, many others work as hired laborers on vessels owned by wealthier members of the community. Shareholders occupy an intermediate position, contributing financially or through labor and sharing in the profits. These structural differences have profound implications for income levels, economic security, and vulnerability to seasonal shocks.

### **1.3 Socio-Economic Vulnerabilities**

Fishermen communities face a unique set of socio-economic vulnerabilities that distinguish them from other rural or semi-urban populations. Among the most significant are:

- **Seasonal Income Fluctuations:** Fishing income is highly seasonal, with periods of high catch during certain months contrasted with lean periods during the monsoon season when fishing is restricted or hazardous. This cyclical income pattern creates chronic cash flow problems for households.
- **Indebtedness:** A large proportion of fishing households are indebted to moneylenders, boat owners, or middlemen. The need to purchase fishing equipment, meet household expenses during lean seasons, and cover social obligations such as weddings or religious functions often forces families into debt cycles.
- **Limited Educational Attainment:** Educational levels within the

fishing community remain low. Many fishermen have only primary-level education or are illiterate, which limits their access to information, formal employment opportunities, and the ability to navigate government schemes.

- **Occupational Risks:** Fishing is one of the most dangerous occupations, with fishermen regularly exposed to rough weather, deep-sea hazards, and the risk of capsizing. Inadequate safety equipment and limited access to distress communication systems exacerbate these risks.
- **Gender Dynamics:** Fishing households are typically male-dominated in terms of direct fishing activity, with women often engaged in fish processing, marketing, and household management. Gender inequities within these households can compound economic vulnerabilities.

#### **1.4 Role of Government Welfare Schemes**

The Government of India and the Government of Tamil Nadu have implemented several welfare schemes targeting the fishermen community. These include insurance schemes such as the Group Accident Insurance Scheme, financial assistance for the purchase of boats and nets, fuel subsidies, housing schemes under schemes like Pradhan Mantri Awas Yojana, and savings and credit facilities through cooperative societies. The Pradhan Mantri Matsya Sampada Yojana (PMMSY) launched in 2020 aims to bring about a Blue Revolution in the fisheries sector through sustainable and responsible development.

Despite these initiatives, awareness and utilization of government schemes remain uneven across fishing communities. Bureaucratic complexities, lack of documentation, geographic isolation, and mistrust of government institutions often prevent eligible beneficiaries from accessing benefits. The present study specifically examines the extent to which fishermen at Colachel Harbor are beneficiaries of government welfare

schemes and their perceptions of government action regarding deep-sea safety.

### **1.5 Significance of the Study**

Understanding the socio-economic profile of the fishermen community is essential for designing targeted policy interventions. Income disparities linked to education and mode of fishing indicate that investment in education and technology adoption can improve economic outcomes. Expenditure patterns reveal the financial burden of loan repayment and housing, suggesting the need for accessible credit and housing support. Safety perceptions and dissatisfaction with government measures highlight gaps in maritime safety policy and implementation. This study provides quantitative evidence that can serve as a baseline for monitoring changes and evaluating the impact of future policy initiatives targeted at the Colachel fishing community.

### **1.6 Limitations of the Study**

The present study has several limitations that should be acknowledged. The sample was drawn exclusively from the Colachel Harbor area, and findings may not be fully generalizable to other fishing communities in Tamil Nadu or India. The cross-sectional design captures a snapshot in time and cannot capture changes in socio-economic conditions over time. Income data, being self-reported, may be subject to recall bias and social desirability effects, particularly for high-income respondents. The exclusion of female family members from the sample limits the study's ability to assess gender dimensions of household economic management. Future longitudinal studies with broader geographic coverage and gender-inclusive sampling would substantially enrich the evidence base.

### **1.7 Objectives of the Study**

Based on the identified research gaps and field observations, the following objectives were established:

1. To analyse the demographic profile of the fishermen in terms of age, gender, marital status, family size, education, and years of fishing experience.
2. To examine whether annual income differs according to education level, fishing experience, and type of fishing activity.
3. To identify the major purpose for which fishermen households spend the largest portion of their income and to analyse the household expenditure pattern.
4. To examine whether perceptions regarding the safety of fishing activities at present differ according to the fishermen's role in fishing activities such as worker, shareholder, and owner.
5. To study the opinion of fishermen regarding the measures taken by the government to help fishermen who are in danger in deep sea.

## **CHAPTER - 2**

### **REVIEW OF LITERATURE**

The primary objective of this chapter is to provide a comprehensive insight into the research problem by presenting theoretical background and reviews from relevant published literature. Studies on socio-economic conditions of fishermen communities, income determinants, government welfare, and safety perceptions have been reviewed to establish the foundation for the current work.

#### **2.1 Socio-Economic Profile of Fishermen in India**

Vivekanandan (2010) documented that the fisheries sector in India supports over 14 million fishers and fish workers, with the coastal fishing community being among the most economically marginalized groups. The study highlighted low educational attainment, seasonal income variability, and high indebtedness as the defining features of fishing households in coastal Tamil Nadu.

Sathiadhas & Narayanakumar (1994) conducted a pioneering study on the economics of artisanal fisheries in India, concluding that the income of artisanal fishermen is considerably lower than that of mechanized fishermen, and that the adoption of improved fishing technology is positively associated



with higher income levels. Their findings remain relevant to the present study context at Colachel, where both traditional and mechanized fishing operations coexist.

Bavinck (2001) examined the social organization of fishing communities in Tamil Nadu and found that occupational roles within fishing particularly the distinction between owners, shareholders, and workers play a decisive role in shaping income levels and economic security. Owners and shareholders were found to have significantly higher economic standing than wage workers, a pattern that the present study also sought to investigate.

## **2.2 Education and Income in Fishing Communities**

Sudarsan (1992) studied the relationship between education and occupational mobility in Indian fishing communities and observed that higher educational attainment was associated with diversification of income sources. However, the study also noted a paradoxical pattern in some communities where illiterate or primary-educated fishermen who were engaged in ownership or shareholding roles earned more than educated counterparts who remained as wage workers.

Kumar & Dey (2012) analysed socio-economic characteristics of fishing households across coastal districts in West Bengal and found a significant inverse relationship between educational level and dependency on fishing as the sole income source. This suggests that education facilitates occupational diversification rather than directly increasing fishing income a nuance that the One-Way ANOVA analysis in the present study was designed to capture. Jentoft & McCay (1995) reviewed studies on fishing co-management and

community governance and emphasized that educational capacity within fishing communities is a key determinant of effective participation in policy processes and resource management. Communities with higher educational profiles were better able to negotiate with government bodies for welfare and safety provisions.

### **2.3 Expenditure Patterns and Indebtedness**

Pillai & Geetha (2005) conducted a detailed study of household expenditure among marine fishermen in Kerala and found that food, loan repayment, and housing expenditures constituted the three largest budget items for fishing households. The study noted that the financial burden of debt servicing often incurred for boat purchases, fishing gear, or medical emergencies was a major constraint on savings and capital accumulation.

Salagrama (2006) examined poverty and livelihoods in the fisheries sector and found that social and religious expenditures, while significant in absolute terms, were often funded through additional borrowing, deepening the debt cycle. The present study's finding that loan repayment (36.0%) is the dominant expenditure category corroborates these earlier observations.

Ramachandra & Srinivasan (2014) highlighted that the shift toward mechanized fishing, while improving catch volumes and income, also significantly increased the operational expenditure on fuel, repairs, and equipment.

## **2.4 Safety, Governance, and Government Schemes**

Bavinck & Gupta (2014) studied fishermen's perceptions of marine governance and safety in South Asia and found widespread dissatisfaction with governmental response to deep-sea dangers. Fishermen across multiple countries reported inadequate rescue infrastructure, insufficient distress communication systems, and a lack of timely government intervention during sea emergencies.

Food and Agriculture Organization (2015), in its Voluntary Guidelines for Small-Scale Fisheries, emphasized that governments have an obligation to ensure the safety and security of small-scale fishermen, including the provision of safety equipment, early warning systems for weather, and rapid response mechanisms for maritime distress. The high proportion of dissatisfied respondents in the present study (97.4%) suggests that these international standards are far from being met in the study area.

Kurien (1992) analysed the impact of state interventions in the fisheries sector in Kerala and argued that while financial subsidies had improved gear and technology access for some fishermen, structural inequalities in access to welfare schemes persisted along lines of boat ownership and caste affiliation. The present study extends this analysis to the Kanyakumari context.

Sivaperuman & Livingstone (2018) conducted a study on government welfare scheme utilization among fishermen in Kanyakumari District and observed a moderate level of scheme uptake despite high eligibility rates, attributing the gap to inadequate awareness campaigns and documentary requirements.

## **2.5 Fishing Safety and Technology Adoption**

Colburn & Clay (2012) argued that the adoption of new fishing technologies including GPS devices, echo sounders, and weather alert systems is strongly associated with perceived improvements in fishing safety and income generation. The present study's finding that the vast majority of respondents had adopted new fishing techniques (mean = 4.86 on a five-point scale) and were equipped with climate information devices (mean = 4.68) is consistent with this broader trend.

Mahon et al. (2013) reviewed studies on fishermen boundary awareness in the Caribbean and South Asia and found that awareness of maritime boundaries through signal devices and border signs was an important safety behavior. The high mean score on fishing boundary awareness (4.55) in the present study suggests that this knowledge is fairly widespread among Colachel fishermen, possibly due to past incidents and community learning.

Allison & Ellis (2001) examined the livelihoods of fishing communities using the sustainable livelihoods framework and concluded that social capital including community networks, cooperative arrangements, and access to information is a critical determinant of resilience in fishing households. The structural variables examined in the present study, including family size, education, and fishing role, can be understood within this broader livelihoods framework. The foregoing review reveals that while a substantial body of literature exists on the socio-economic conditions of fishing communities in India, detailed quantitative studies focused on specific harbor-based communities in Kanyakumari District particularly

examining the joint effects of education, fishing mode, and role on income, along with expenditure patterns and government welfare perceptions remain limited. The present study addresses this gap by providing primary data-based evidence from 114 respondents at Colachel Fishing Harbor, employing rigorous statistical techniques to derive actionable insights.

## **CHAPTER - 3**

### **DATA DESCRIPTION AND METHODOLOGY**

This chapter outlines the methodological framework adopted for the present study. It covers the study area, sampling design, data collection techniques, variables of interest, and the statistical tools employed for analysis. A well-defined methodology is essential for ensuring the reliability and validity of the research findings.

#### **3.1 Study Area**

The study was conducted at Colachel Fishing Harbor, located in Kanyakumari District, Tamil Nadu, India. Colachel is situated at approximately 8.17°N latitude and 77.25°E longitude, on the southwestern tip of the Indian peninsula. The harbor is one of the major fishing centers in South Tamil Nadu and serves as the operational base for hundreds of fishing vessels ranging from traditional country boats to large mechanized trawlers.

The fishing community in Colachel and its surrounding areas has a long historical presence and is predominantly composed of families whose livelihoods have depended on marine fishing for generations. The community is exposed to the seasonal fishing cycle governed by the monsoon, with peak fishing activity occurring during the post-monsoon and summer months and

restricted activity during the southwest monsoon season (June to August), which corresponds to the official fishing ban period mandated by the Tamil Nadu government.

### **3.2 Study Design**

The study adopted a cross-sectional descriptive design using primary data. Data were collected at a single point in time from a representative sample of fishermen residing in and around the Colachel Fishing Harbor area. The cross-sectional design is appropriate for this study as it allows for the assessment of the current socio-economic status of the community and the examination of associations between demographic, economic, and perceptual variables without the complexity of longitudinal tracking.

### **3.3 Sampling Technique**

A purposive sampling technique was employed to identify respondents who were actively engaged in fishing activities associated with Colachel Harbor. The inclusion criteria required that respondents be currently engaged in fishing as their primary occupation or as boat owners or shareholders associated with the harbor. Retired fishermen, fish vendors, and individuals engaged solely in fish processing were excluded from the study.

A total of 114 respondents were successfully surveyed. The sample size was determined based on practical feasibility, the homogeneity of the target population, and established practices in socio-economic surveys of similar communities. The respondents were drawn from different fishing settlements

and landing centers within the operational area of Colachel Harbor to ensure geographic representation within the community.

### **3.4 Data Collection**

Primary data were collected through a structured questionnaire administered via personal interviews. Personal interviews were preferred over self-administered questionnaires due to the relatively low literacy levels in the target population. Trained enumerators fluent in Tamil conducted the interviews at the homes and landing sites of the respondents, ensuring that all questions were clearly understood and accurately answered.

The questionnaire was designed in English and translated into Tamil for field administration. It covered the following broad thematic areas:

- Demographic information: age, gender, marital status, education, family size, number of educated and employed family members.
- Occupational profile: fishing role, income type, mode of fishing, years of experience.
- Economic characteristics: annual income, economic status, monthly expenditure, savings.
- Household expenditure priorities: major purpose of household spending.
- Perceptions and attitudes: Likert-scale items on income stability, adoption of new techniques, climate information readiness, fishing safety, boundary awareness.



- Government welfare: access to government schemes and perception of government action on deep-sea safety.

### **3.5 Data Description**

The study utilized 25 variables (excluding the open-ended suggestion item), covering nominal, ordinal, and ratio scales of measurement. The primary variables of interest were:

- Dependent Variable: Annual Income (Ratio scale), used in ANOVA analyses.
- Independent Variables: Educational Qualification (Ordinal), Fishing Role (Nominal), Fishing Mode (Nominal), Experience (Ratio), Economic Status (Ordinal).
- Likert-Scale Variables (Ordinal, 15): Income stability in normal season, income stability in off-season, adoption of new techniques, climate information readiness, fishing safety perception, and fishing boundary awareness.
- Nominal Variables: Gender, Marital Status, Savings status, Government Welfare access, Government Steps perception.

### **3.6 Ethical Considerations**

All participants were informed about the nature and purpose of the study prior to data collection. Participation was entirely voluntary, and respondents were assured of confidentiality. No personal identifiers such as names or contact numbers were recorded in the dataset. The data were stored securely

and used solely for the purposes of this academic project. Verbal informed consent was obtained from each respondent before the interview commenced.

### 3.7 Statistical Methods

#### 3.7.1 Descriptive Statistics

Frequencies, percentages, mean, standard deviation, minimum, and maximum values were computed for all relevant variables to describe the demographic and socio-economic profile of the sample.

*Mean:*

$$\bar{X} = \frac{\sum X_i}{n}$$

*Standard Deviation:*

$$\sigma = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$$

#### 3.7.2 One-Way Analysis of Variance (ANOVA)

This is used to test whether mean annual income differs significantly across categories of educational qualification, fishing experience groups, and mode of fishing. Levene's test was applied to check homogeneity of variances.

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

$$F = \frac{\frac{\sum n_i(\bar{X}_i - \bar{X})^2}{k-1}}{\frac{\sum (X_{ij} - \bar{X}_i)^2}{N-k}}$$

**Levene's Test:**

$$W = \frac{(N - k)}{(k - 1)} \cdot \frac{\sum k(Z_i - Z)^2}{\sum (Z_{ij} - Z_i)^2}$$

**Assumptions:**

- Observations are independent.
- The dependent variable is normally distributed within each group.
- Homogeneity of variances (equal variances cross groups).
- Groups are independent of each other.
- Dependent variable is measured at interval or ratio scale.

**3.7.3 Multiple Linear Regression Analysis**

This technique is used to examine the effect of multiple independent variables such as educational qualification, fishing experience, and role in fishing activity on the dependent variable, annual income. It helps in identifying the significant predictors and measuring their impact on income.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

Where,

$Y$  = Annual Income

$X_1$  = Educational Qualification

$X_2$  = Fishing Experience

$X_3$  = Role in Fishing Activity

$\beta_0$  = Intercept and  $\beta_1, \beta_2, \beta_3$  = Regression Coefficients

$\epsilon$  = Error Term

**Assumptions:**

- Linearity: There is a linear relationship between dependent and independent variables.
- Independence: Observations are independent of each other.
- Homoscedasticity: Constant variance of residuals.
- Normality: Residuals are normally distributed.
- No Multicollinearity: Independent variables are not highly correlated.

**3.7.4 Chi-Square Test**

This is used to examine the association between the fishermen's role (worker, shareholder, owner) and their perception of fishing safety. Pearson's Chi-Square statistic was computed to assess statistical significance.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

**Expected Frequency:**

$$E = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand Total}}$$

**Assumptions:**

- Data is categorical (nominal or ordinal).
- Observations are independent.
- Expected frequency in each cell should be at least 5.

- Categories are mutually exclusive.
- Sample size should be sufficiently large

### **3.7.5 Frequency**

Frequency is the count of how often a specific value, response, or category appears in a dataset. It helps in understanding the distribution and pattern of data by showing which values occur more or less frequently.

$$f_i = \text{Number of times the } i^{th} \text{ value occurs}$$

## **3.8 Tools for Data Analysis**

The collected data were coded, cleaned, and entered into SPSS (Statistical Package for Social Sciences) Version 22.0 for analysis. The following statistical tools were employed.

## **CHAPTER - 4**

### **DATA ANALYSIS AND INTERPRETATION**

This chapter presents the results of various statistical analyses conducted to address the objectives of the study. The findings are organized in accordance with the five research objectives, moving from demographic profiling to income analysis, expenditure patterns, safety perceptions, and government welfare opinions.

#### **4.1 Demographic Profile of the Fishermen**

The first objective was to analyse the demographic profile of the fishermen in terms of age, gender, marital status, family size, education, and years of fishing experience.

##### **4.1.1 Gender and Marital Status**

All 114 respondents (100) in the study were male. This finding is consistent with the broader pattern of gender division in marine fishing activities in India, where fishing at sea is almost exclusively carried out by men. Women in fishing communities typically engage in shore-based activities such as fish vending, processing, and net repair. Similarly, all 114 respondents (100) were married, indicating that every fisherman in the

sample carried household responsibilities, which has direct implications for expenditure patterns and financial decision-making.

Table 4.1: Gender Distribution of Respondents

| <b>Gender</b> | <b>Frequency</b> | <b>Percent</b> | <b>Valid Percent</b> |
|---------------|------------------|----------------|----------------------|
| Male          | 114              | 100.0          | 100.0                |

In Table 4.1, All 114 respondents (100) in the study were male. This finding is consistent with the broader pattern of gender division in marine fishing activities in India, where fishing at sea is almost exclusively carried out by men. Women in fishing communities typically engage in shore-based activities such as fish vending, processing, and net repair.

Table 4.2: Marital Status of Respondents

| <b>Marital Status</b> | <b>Frequency</b> | <b>Percent</b> | <b>Valid Percent</b> |
|-----------------------|------------------|----------------|----------------------|
| Married               | 114              | 100.0          | 100.0                |

In Table 4.2, all 114 respondents (100) were married, indicating that every fisherman in the sample carried household responsibilities, which has direct implications for expenditure patterns and financial decision-making.

#### **4.1.2 Educational Qualification**

The educational profile of the respondents reveals a predominantly low level of formal education within the fishing community. As presented in Table 4.2, 74.6 of respondents (n = 85) had completed only primary education, making this the largest educational category. A further 16.7 (n = 19) had secondary education, while 7.0 (n = 8) were illiterate. Only 1.8 (n = 2) had attained higher secondary education, representing a negligible proportion of the sample. This pattern highlights the chronic educational

disadvantage prevalent in coastal fishing communities and its potential impact on economic mobility and access to government programs.

Table 4.3: Annual Income by Role in Fishing Activity

| Source         | Sum of Squares | df  | Mean Square | F     | Sig.  |
|----------------|----------------|-----|-------------|-------|-------|
| Between Groups | 3855           | 17  | 2267        | 3.078 | 0.000 |
| Within Groups  | 7073           | 96  | 7368        |       |       |
| Total          | 10928          | 113 |             |       |       |

Table 4.3 shows that annual income differs significantly based on fishing experience ( $F = 3.078$ ,  $p < 0.05$ ). Since the p-value is less than 0.05, the null hypothesis is rejected, indicating that fishing experience has a significant effect on annual income.

#### 4.1.3 Age, Family Size, and Experience

Table 4.4, presents the descriptive statistics for the continuous demographic variables. The age of respondents ranged from 28 to 86 years, with a mean of 51.20 years ( $SD = 10.64$ ), indicating that the fishing workforce at Colachel is largely composed of middle-aged and older individuals. The relatively high mean age raises important questions about workforce renewal and the succession of fishing knowledge to younger generations. Family size ranged from 2 to 9 members, with a mean of 4.57 members ( $SD = 1.09$ ), which is typical of semi-rural Tamil Nadu households. The years of fishing experience ranged from 5 to 70 years, with a mean of 36.18 years ( $SD = 12.27$ ). This high average experience level underscores the



deep intergenerational roots of fishing as an occupation in the community. Fishermen typically begin their occupation early in life, often during their teenage years, and continue throughout their working lives.

Table 4.4: Descriptive statistics of Demographic Profile

| <b>Variable</b>             | <b>N</b> | <b>Min</b> | <b>Max</b> | <b>Mean</b> | <b>Std. Dev</b> |
|-----------------------------|----------|------------|------------|-------------|-----------------|
| Age (years)                 | 114      | 28         | 86         | 51.20       | 10.640          |
| Number of Family Members    | 114      | 2          | 9          | 4.57        | 1.087           |
| Years of Fishing Experience | 114      | 5          | 70         | 36.18       | 12.271          |

Table 4.4 shows that the respondents have an average age of 51.20 years, with ages ranging from 28 to 86 years. The average family size is 4.57 members, indicating medium-sized households. The mean fishing experience is 36.18 years, suggesting that most respondents are highly experienced in fishing.

## **4.2 One-Way ANOVA**

The second objective examined whether annual income differs significantly according to educational qualification, fishing role, and mode of fishing. One-Way ANOVA was employed for this purpose.

### **4.2.1 Annual Income by Educational Qualification**

Before conducting ANOVA, Levene's test for homogeneity of variances was applied. The result (Levene Statistic = 7.758, df1 = 3, df2 = 110,  $p = 0.000$ ) indicated that the assumption of equal variances was violated. Nevertheless, ANOVA was conducted as it is robust to violations of homogeneity when group sizes are reasonably large, and results are interpreted with

appropriate caution. Table 4.4 presents the mean annual income across educational groups. Illiterate fishermen reported the highest mean annual income (Rs. 1,987,500), followed by primary-educated fishermen (Rs. 844,705), secondary-educated (Rs. 521,053), and higher-secondary educated fishermen (Rs. 325,000). The ANOVA result yielded  $F(3, 110) = 4.927$ ,  $p = 0.003$ , indicating a statistically significant difference in mean annual income across educational groups.

#### *Hypothesis*

Null Hypothesis ( $H_0$ ): There is no significant difference in annual income among different groups of fishermen (based on education / fishing experience / role).

Alternative Hypothesis ( $H_1$ ): There is a significant difference in annual income among different groups of fishermen.

Table 4.5: Education Level and Mean Income of Respondents

| <b>Education Level</b> | <b>N</b> | <b>Mean</b> | <b>Std. Dev</b> | <b>F-value</b> | <b>p-value</b> |
|------------------------|----------|-------------|-----------------|----------------|----------------|
| Illiterate             | 8        | 19,87,500   | 18,85,612       | 4.927          | 0.003*         |
| Primary                | 85       | 8,44,706    | 9,18,138        |                |                |
| Secondary              | 19       | 5,21,053    | 1,87,317        |                |                |
| Higher Secondary       | 2        | 3,25,000    | 1,06,066        |                |                |
| Total                  | 114      | 8,61,842    | 9,83,431        |                |                |

Table 4.5 shows that mean annual income differs across education levels, with illiterate respondents having the highest income (19,87,500) and higher secondary respondents having the lowest (3,25,000). The ANOVA result ( $F = 4.927$ ,  $p = 0.003$ ) indicates that this difference is statistically significant,

implying that income varies significantly with educational qualification.

#### 4.2.2 Annual Income by Fishing Mode

The analysis also examined income differences between traditional and mechanized fishermen. Mechanized fishermen (n = 89) reported a significantly higher mean income (Rs. 773,337) compared to traditional fishermen (n = 25) who reported a mean income of Rs. 397,200. Table 4.6 This finding is consistent with the broader literature indicating that mechanized fishing yields larger catches and consequently higher income, albeit at the cost of higher operational expenditure.

##### *Hypothesis*

Null Hypothesis ( $H_0$ ): There is no significant difference in annual income among different groups of fishermen (based on education / fishing experience / role).

Alternative Hypothesis ( $H_1$ ): There is a significant difference in annual income among different groups of fishermen.

Table 4.6: Income Distribution by Fishing Mode

| <b>Fishing Mode</b> | <b>N</b> | <b>Mean Income Rs.</b> | <b>Std. Deviation</b> |
|---------------------|----------|------------------------|-----------------------|
| Traditional         | 25       | 3,97,200               | 2,63,480              |
| Mechanized          | 89       | 7,73,337               | 10,84,521             |
| Total               | 114      | 6,88,622               | 9,53,200              |

Table 4.6, shows that there is a highly significant difference in annual income based on the role in fishing activity ( $F = 44.984$ ,  $p = 0.000$ ). Since the p-value is less than 0.05, the null hypothesis is rejected, indicating that the

role in fishing activity has a strong influence on annual income.

#### 4.2.3 Annual Income by Role in Fishing Activity

Among the 114 respondents, 66 (57.9) were workers, 34 (29.8) were owners, and 14 (12.3) were shareholders. The dominance of wage workers in the sample reflects the typical hierarchical structure of fishing enterprises in which ownership is concentrated among a smaller number of individuals. Table 4.6 presents this distribution.

##### *Hypothesis*

Null Hypothesis ( $H_0$ ): There is no significant difference in annual income among different groups of fishermen (based on education / fishing experience / role).

Alternative Hypothesis ( $H_1$ ): There is a significant difference in annual income among different groups of fishermen.

Table 4.7: Annual income by fishing activity

| Source         | Sum of Squares | df  | Mean Square | F      | Sig.  |
|----------------|----------------|-----|-------------|--------|-------|
| Between Groups | 4892           | 2   | 2446        | 44.984 | 0.000 |
| Within Groups  | 6036           | 111 | 5438        |        |       |
| Total          | 10928          | 113 |             |        |       |

Table 4.7, shows that annual income differs significantly based on the role in fishing activity ( $F = 44.984$ ,  $p < 0.05$ ). This means the role of the fisherman

has a strong effect on their income.

### 4.3 Multiple Regression Analysis of Annual Income

The fourth objective was to examine the effect of educational qualification, fishing experience, and role in fishing activity on the annual income of fishermen.

Table 4.8: ANOVA Results for Regression Model

| Source     | Sum of Squares | df  | Mean Square | F      | Sig.  |
|------------|----------------|-----|-------------|--------|-------|
| Regression | 2670           | 3   | 8901        | 11.856 | 0.000 |
| Residual   | 8258           | 110 | 7507        |        |       |
| Total      | 10928          | 113 |             |        |       |

In Table 4.8, The ANOVA results ( $F = 11.856$ ,  $p < 0.05$ ) confirm that the regression model is statistically significant, indicating that the independent variables jointly have a significant effect on annual income.

Table 4.9: Regression Coefficients

| Variables                 | B     | Std. Error | Beta   | t      | Sig.  |
|---------------------------|-------|------------|--------|--------|-------|
| Constant                  | -5491 | 7482       |        | -0.734 | 0.465 |
| Fishing Experience        | 2875  | 9523       | 0.359  | 3.019  | 0.003 |
| Role in Fishing Activity  | 3069  | 91432.768  | 0.280  | 3.358  | 0.001 |
| Educational Qualification | -7104 | 2154       | -0.039 | -0.330 | 0.742 |

Table 4.9 presents the results of the multiple regression analysis. The model summary indicates that the R value is 0.494, showing a moderate relationship between the dependent and independent variables. The R square value of 0.244 implies that 24.4% of the variation in annual income is explained by educational qualification, fishing experience, and role in fishing

activity.

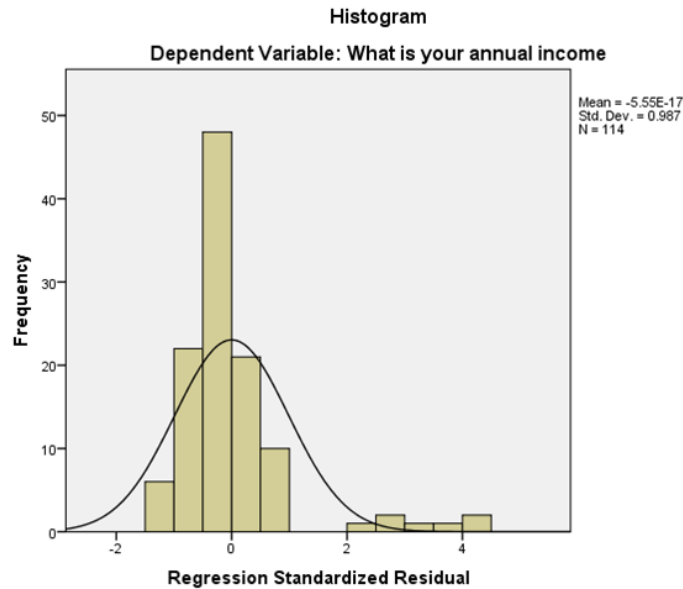


Figure 4.1: Histogram of standardized residuals showing approximate normal distribution

The histogram figure:4.1 of standardized residuals. The distribution appears approximately normal, indicating that the normality assumption is satisfied.

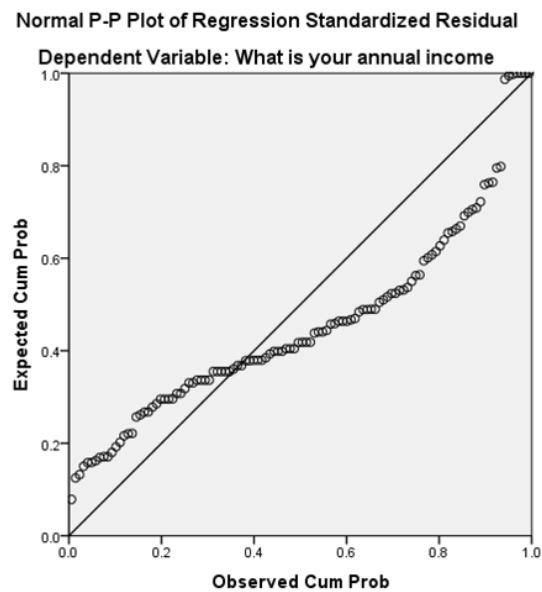


Figure 4.2: Normal P-P plot of standardized residuals indicating normality assumption

The Normal P-P plot of standardized residuals. The points lie close to the diagonal line, indicating that the residuals are normally distributed.

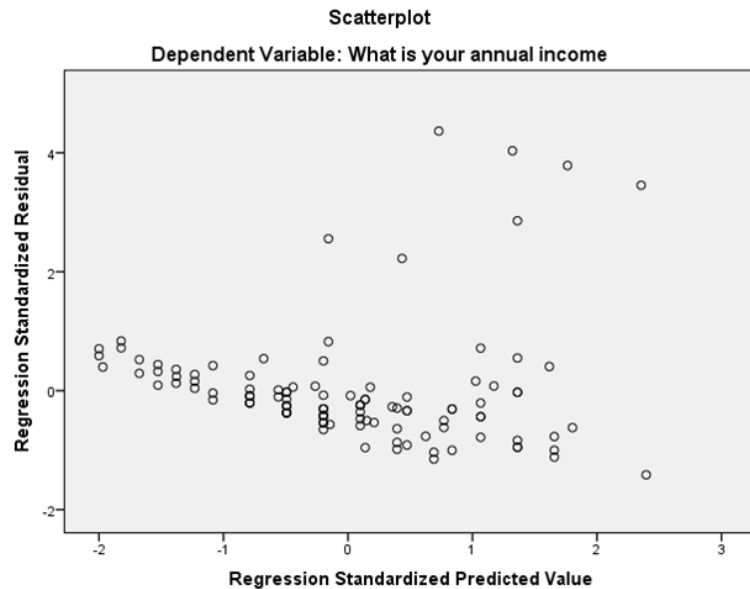


Figure 4.3: Scatter plot of standardized residuals versus predicted values showing homoscedasticity

The scatter plot of standardized residuals versus predicted values. The points are randomly scattered without any clear pattern, indicating that the assumption of homoscedasticity

### *Hypothesis*

Null Hypothesis ( $H_0$ ): Educational qualification, fishing experience, and role in fishing activity do not have a significant effect on annual income.

Alternative Hypothesis ( $H_1$ ): At least one of the variables (educational qualification, fishing experience, or role in fishing activity) has a significant effect on annual income.

#### 4.4 Household Expenditure Pattern

The third objective was to identify the major purpose for which fishermen households spend the largest portion of their income and to analyse the household expenditure pattern.

Table 4.10: Expenditure Distribution of Respondents

| <b>Expenditure Category</b>         | <b>Frequency</b> | <b>Percent (%)</b> |
|-------------------------------------|------------------|--------------------|
| Loan Repayment                      | 41               | 36.0               |
| House Rent/Maintenance              | 33               | 28.9               |
| Fishing Expenses (Fuel, Nets, Boat) | 24               | 21.1               |
| Social/Religious Functions          | 9                | 7.9                |
| Health/Medical Expenses             | 4                | 3.5                |
| Education of Children               | 2                | 1.8                |
| Food and Groceries                  | 1                | 0.9                |
| Total                               | 114              | 100.0              |

Table 4.10 presents the frequency distribution of the primary expenditure category as reported by respondents. Loan repayment (36.0, n = 41) emerged as the single largest expenditure category, underscoring the severe indebtedness prevalent in the fishing community. House rent and maintenance (28.9, n = 33) was the second most common primary expenditure, followed by fishing expenses including fuel, nets, and boat maintenance (21.1, n = 24). Social or religious functions accounted for 7.9 (n = 9), while health and medical expenses (3.5, n = 4), education of children (1.8, n = 2), and food and groceries (0.9, n = 1) were cited by smaller proportions.

Figure 4.1 The relatively low priority assigned to food expenditures may



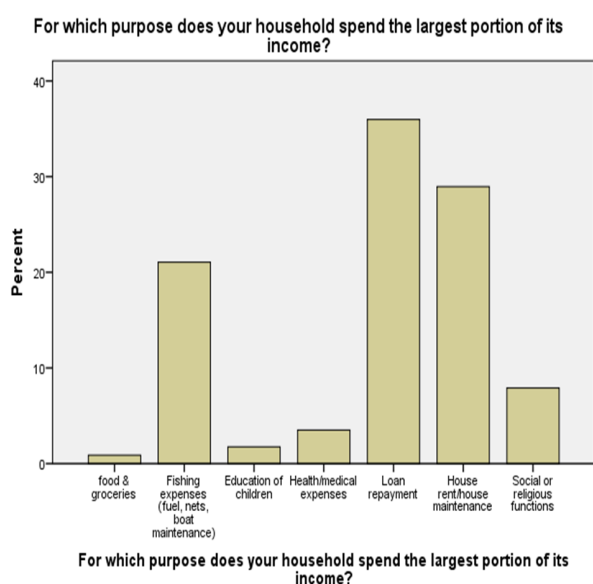


Figure 4.4: Household Income Expenditure Pattern

suggest that households have access to self-sourced fish protein, reducing food expenditure relative to other budget items.

#### 4.4.1 Savings and Income Stability

From the 114 respondents, 72 (63.2) reported having savings or investments, while 42 (36.8) had no savings. While the majority reported having some savings, the amount and form of savings were not captured in the current study. Income stability was assessed through two Likert-scale items. The mean score for income stability during the normal season was 4.71 (out of 5), indicating high reported stability during peak fishing periods. However, the mean score for income stability during the off-season dropped sharply to 1.47, reflecting the extreme income volatility experienced during lean months. This finding has critical policy implications for the design of social protection mechanisms for fishing households.

Table 4.11 shows that respondents strongly agree that their income is

Table 4.11: Income Stability Analysis

| <b>Variable</b>               | <b>Mean</b> | <b>Std. Deviation</b> | <b>Interpretation</b> |
|-------------------------------|-------------|-----------------------|-----------------------|
| Income stable – Normal Season | 4.71        | 0.72                  | Strongly Agree        |
| Income stable – Off Season    | 1.47        | 0.78                  | Strongly Disagree     |

stable during the normal season (mean = 4.71), while they strongly disagree that income is stable during the off-season (mean = 1.47). This indicates that fishermens income is stable only during the normal season and becomes unstable during the off-season.

#### **4.5 Safety Preceptions by Fishing Role**

The fourth objective examined whether perceptions of fishing safety differ according to the fishermen's role (worker, shareholder, owner). A crosstabulation was performed between the fishing role variable and the Likert-scale item 'Fishing is safer now than it was in the past.' The results are presented in Table 4.9

The analysis provides a comparative view of how safety perceptions vary across different categories of fishermen. By examining the distribution of responses, it becomes possible to identify whether certain groups perceive improvements in safety more positively than others. This helps in understanding the influence of occupational role on risk perception within the fishing community.

The Table 4.12, The crosstabulation reveals that a large majority of respondents across all three roles gave a neutral response to the question of whether fishing is safer now than in the past. Workers (87.9), shareholders (78.6), and owners (85.3) predominantly selected the neutral option,

Table 4.12: Opinion by Fishing Role

| <b>Fishing Role</b> | <b>Disagree (%)</b> | <b>Neutral (%)</b> | <b>Agree (%)</b> | <b>Strongly Agree (%)</b> | <b>Total (N)</b> |
|---------------------|---------------------|--------------------|------------------|---------------------------|------------------|
| Worker              | 0.0                 | 87.9               | 7.6              | 4.5                       | 66               |
| Shareholder         | 0.0                 | 78.6               | 7.1              | 14.3                      | 14               |
| Owner               | 5.9                 | 85.3               | 0.0              | 8.8                       | 34               |
| Total               | 1.8                 | 86.0               | 5.3              | 7.0                       | 114              |

suggesting widespread ambivalence about improvements in fishing safety. Very few respondents in any category expressed strong agreement that fishing had become safer. A small proportion of owners (5.9) actively disagreed with the statement, reflecting perhaps greater awareness of the risks associated with deep-sea fishing due to greater operational exposure.

#### **4.5.1 New Techniques and Climate Information Readiness**

Two additional Likert-scale variables examined technological adaptation within the fishing community. The mean score for adoption of new fishing techniques was 4.86 (out of 5), Table 4.10 indicating near-universal adoption of improved fishing methods. The mean score for climate information readiness (access to devices for receiving climate, storm, and location information) was 4.68, reflecting high penetration of communication and navigation equipment. Similarly, fishing boundary awareness showed a high mean of 4.55, indicating that most fishermen are aware of territorial fishing boundaries through signal devices or border signs.

Table 4.13 shows that most respondents strongly agree that they have adopted new fishing techniques (mean = 4.86), are equipped with climate information devices (mean = 4.68), and are aware of fishing boundaries (mean = 4.55). However, their opinion on whether fishing is safer now than in the

Table 4.13: Adoption and Awareness of Fishing Practices

| Variable                           | N   | Mean | Std. Deviation | Decision       |
|------------------------------------|-----|------|----------------|----------------|
| Adopted New Fishing Techniques     | 114 | 4.86 | 0.42           | Strongly Agree |
| Equipped with Climate Info Devices | 114 | 4.68 | 0.71           | Strongly Agree |
| Aware of Fishing Boundaries        | 114 | 4.55 | 0.89           | Strongly Agree |
| Fishing is Safer Now than Past     | 114 | 3.18 | 0.53           | Neutral        |

past is neutral (mean = 3.18). This indicates high awareness and adoption of modern practices, but uncertainty regarding safety improvements.

## **4.6 Government Welfare and Deep-Sea Safety Perceptions**

The fifth objective was to study the opinion of fishermen regarding the measures taken by the government to help fishermen who are in danger in deep sea. Two questions addressed this objective: whether respondents received government welfare schemes, and whether they believed the government was taking adequate steps to help fishermen in deep-sea danger.

### **4.6.1 Access to Government Welfare Schemes**

Of the 114 respondents, 61 (53.5) reported that they received government welfare schemes, while 53 (46.5) indicated that they did not. Although slightly over half the respondents were beneficiaries, the high proportion of non-beneficiaries (nearly one in two) points to significant gaps in welfare coverage within the community. In Table 4.11, This may be attributable to lack of awareness, documentary requirements, or exclusion from eligibility due to informal employment arrangements.

Table 4.14 shows that respondents strongly agree that they have adopted new fishing techniques (mean = 4.86), are equipped with climate information

Table 4.14: Welfare Receipt Status

| <b>Category</b>               | <b>Frequency</b> | <b>Percent (%)</b> |
|-------------------------------|------------------|--------------------|
| Yes – Receives Welfare        | 61               | 53.5               |
| No – Does Not Receive Welfare | 53               | 46.5               |
| Total                         | 114              | 100.0              |

devices (mean = 4.68), and are aware of fishing boundaries (mean = 4.55). However, they remain neutral about whether fishing is safer now than in the past (mean = 3.18), indicating uncertainty regarding safety despite high awareness and adoption of modern practices.

#### **4.6.2 Perception of Government Steps for Deep-Sea Safety**

The results regarding fishermen's perceptions of government action on deep-sea safety are striking. An overwhelming 97.4 (n = 111) of respondents felt that the government is not taking adequate steps to help fishermen who are in danger at sea. Only 2.6 (n = 3) expressed the view that the government was taking adequate steps. This near-unanimous dissatisfaction with government response to maritime safety emergencies represents one of the most critical findings of the present study.

Table 4.15: Steps taken for welfare

| <b>Category</b>       | <b>Frequency</b> | <b>Percent (%)</b> |
|-----------------------|------------------|--------------------|
| Yes – Taking Steps    | 3                | 2.6                |
| No – Not Taking Steps | 111              | 97.4               |
| Total                 | 114              | 100.0              |

This finding is particularly alarming given that deep-sea fishing carries significant life risks, and inadequate rescue infrastructure can result in

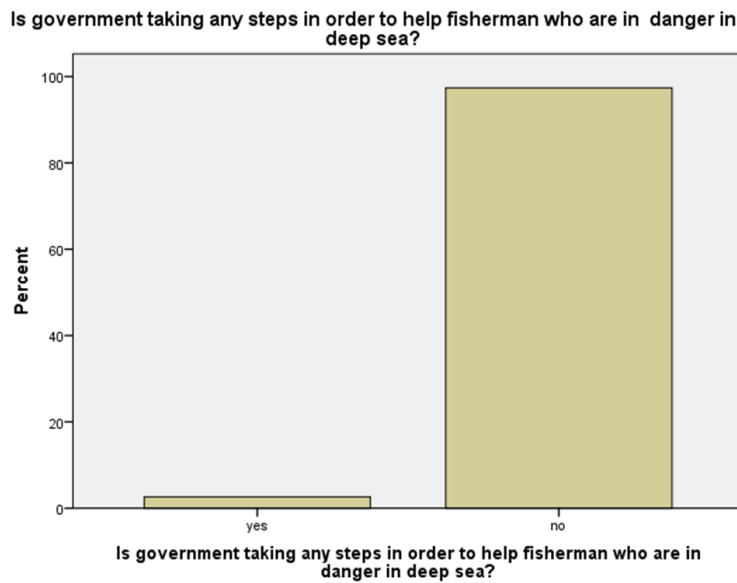


Figure 4.5: Fishermen's Perception of Government Support for Deep-Sea Safety

preventable fatalities during storms, equipment failures.

Figure 4.5 shows that the majority of fishermen believe that the government is not taking adequate steps to support those in danger in deep-sea fishing, while only a very small proportion feel that sufficient measures are being taken. This indicates a high level of dissatisfaction among fishermen regarding government support for deep-sea safety.

#### 4.6.3 Suggested Government Measures

The open-ended question asking respondents to suggest government measures yielded a recurring theme: the establishment of dedicated safety and helpline sectors to assist fishermen during dangerous situations at sea. Other suggestions included provision of modern communication devices, improved coast guard presence, compensation for families of fishermen lost at sea, regulated insurance coverage, and better enforcement of

weather-based fishing restrictions. These qualitative suggestions complement the quantitative findings and provide actionable recommendations for policy-makers.

#### 4.7 Self-Reported Economic Status

The self-reported economic status of the respondents further contextualizes the income and expenditure findings. As shown in Table 4.13, the majority (49.1,  $n = 56$ ) classified themselves as belonging to the middle economic category, followed by 29.8 ( $n = 34$ ) who identified as lower middle class. A smaller proportion identified as upper middle class (11.4,  $n = 13$ ) or high economic status (9.6,  $n = 11$ ). These self-assessments are broadly consistent with the income data and reflect the heterogeneity in economic standing within the fishing community.

Table 4.16: Economic Status Distribution

| <b>Economic Status</b> | <b>Frequency</b> | <b>Percent (%)</b> |
|------------------------|------------------|--------------------|
| Lower Middle           | 34               | 29.8               |
| Middle                 | 56               | 49.1               |
| Upper Middle           | 13               | 11.4               |
| High                   | 11               | 9.6                |
| Total                  | 114              | 100.0              |

Table 4.16 shows that nearly half of the respondents (49.1%) belong to the middle economic group, followed by 29.8% in the lower middle category. A smaller proportion falls under upper middle (11.4%) and high (9.6%) categories. This indicates that most fishermen belong to the middle and lower middle economic groups.

## **CHAPTER - 5**

### **SUMMARY AND CONCLUSION**

This chapter presents a consolidated summary of the major findings from the cross-sectional socio-economic study of fishermen in and around Colachel Fishing Harbor, Kanyakumari District, Tamil Nadu. The study was based on 114 respondents and examined five key areas: demographic profile, income variation, expenditure pattern, fishing safety perceptions, and opinions on government welfare measures.

#### **5.1 Summary of Results**

The findings indicate that the fishing workforce is exclusively male and predominantly married, reflecting strong occupational and cultural traditions within the community. The average age of the respondents is relatively high, suggesting an ageing workforce with extensive fishing experience. Educational attainment is low, with the majority having only primary education and a small proportion being illiterate. The average family size indicates moderate household dependency. Overall, the demographic profile reflects a mature and experienced community with limited access to formal education.

The analysis of income variation shows that income differs significantly



across groups. However, the relationship between education and income is not straightforward. In many cases, individuals with lower educational levels earn higher income due to their ownership or shareholding roles in fishing activities. In contrast, relatively educated individuals are often engaged as workers with lower income levels. Additionally, mechanized fishing yields higher income compared to traditional fishing, highlighting the importance of technology in improving economic outcomes.

The study also reveals that loan repayment is the major component of household expenditure, followed by housing and fishing-related operational expenses. This indicates a high level of indebtedness among fishermen households. Income stability varies significantly between seasons, with relatively stable earnings during the normal season and considerable instability during the off-season. This seasonal variation reflects the vulnerability of fishing livelihoods.

The analysis of fishing safety perceptions shows that opinions are largely similar across different occupational roles, with no significant variation observed. Although fishermen have adopted modern techniques and safety-related technologies, these advancements have not led to a strong perception of improved safety conditions.

With regard to government welfare measures, the study finds that while some fishermen benefit from existing schemes, a substantial proportion do not receive such support. A majority of respondents expressed dissatisfaction with the governments efforts to ensure safety in deep-sea fishing. This indicates a gap between policy implementation and the actual needs of the fishing community.

## **5.2 Conclusion**

The study highlights that the fishermen community in Colachel is characterized by high occupational experience but low educational attainment, along with significant economic vulnerability. Income levels are influenced more by occupational role and type of fishing than by education alone. The dominance of loan repayment in household expenditure reflects persistent financial stress among fishermen.

Seasonal income instability further exacerbates economic insecurity, indicating the need for targeted support during off-seasons. Despite the adoption of modern fishing techniques, safety concerns remain largely unaddressed, pointing to the limitations of technology without adequate institutional support.

The findings also reveal considerable dissatisfaction with government welfare measures, particularly in relation to deep-sea safety. This underscores the need for improved policy implementation, better awareness, and stronger support mechanisms for fishermen.

Overall, the study emphasizes the need for comprehensive interventions focusing on education, financial stability, technological support, and effective governance to improve the socio-economic conditions and safety of the fishing community.

## **5.3 Policy Recommendations**

- Expand and streamline government welfare scheme delivery to eliminate the 46.5% coverage gap, including door-step registration

drives and digital documentation support.

- Establish dedicated deep-sea distress helplines and emergency response units at major fishing harbors including Colachel, with multi-lingual support and 24/7 availability.
- Design off-season income support mechanisms such as fishing ban compensation, diversified livelihood training, and micro-enterprise support to reduce the cyclical income vulnerability documented in this study.
- Facilitate access to formal credit at regulated interest rates to reduce dependence on informal moneylenders, which is the primary driver of the debt burden reflected in the expenditure pattern.
- Invest in adult literacy and vocational training programs for fishermen and their family members to gradually improve the community's human capital base and expand livelihood options.
- Conduct regular community-level awareness campaigns about existing government schemes, safety protocols, and fishing boundary regulations in the local language (Tamil & malayalam) to improve uptake and compliance.

## References

- Allison, E. H., & Ellis, F. (2001). The livelihoods approach and management of small-scale fisheries. *Marine Policy*, 25(5), 377–388.
- Bavinck, M. (2001). *Marine resource management: Conflict and regulation in the fisheries of the coromandel coast*. Sage Publications.
- Bavinck, M., & Gupta, J. (2014). Fisheries governance and safety perceptions in south asia. *Ocean and Coastal Management*.
- Colburn, L. L., & Clay, P. M. (2012). Adoption of technology and fishing safety. *Marine Policy*.
- Food and Agriculture Organization. (2015). *Voluntary guidelines for securing sustainable small-scale fisheries* (Tech. Rep.). FAO.
- Jentoft, S., & McCay, B. J. (1995). User participation in fisheries management: Lessons from international experiences. *Marine Policy*.
- Kumar, P., & Dey, M. M. (2012). Socio-economic characteristics of marine fishing households in india. *Agricultural Economics Research Review*.
- Kurien, J. (1992). State interventions and fisheries development in kerala. *Economic and Political Weekly*.
- Mahon, R., McConney, P., & Roy, R. N. (2013). Awareness of fishing boundaries and safety practices. *Ocean and Coastal Management*.
- Pillai, N. G. K., & Geetha, R. (2005). Household expenditure patterns of marine fishermen in kerala. *Fishery Technology*.

## REFERENCES

---

- Ramachandra, T. V., & Srinivasan, U. (2014). Mechanized fishing and its economic implications. *Journal of Environmental Studies*.
- Salagrama, V. (2006). *Trends in poverty and livelihoods in coastal fishing communities of india*. FAO.
- Sathiadhas, R., & Narayanakumar, R. (1994). Economics of artisanal fisheries in india. *Marine Fisheries Information Service*.
- Sivaperuman, C., & Livingstone, C. (2018). Utilization of government welfare schemes among fishermen in kanyakumari district. *International Journal of Fisheries and Aquatic Studies*.
- Sudarsan, R. (1992). Education and occupational mobility in fishing communities. *Economic and Political Weekly*.
- Vivekanandan, E. (2010). *Socio-economic status of marine fishers in india*. CMFRI.

## **APPENDIX**

### **SOCIO-ECONOMIC STUDY OF FISHERMAN COMMUNITY AROUND COLACHEL FISHING HARBOUR**

#### **QUESTIONNAIRE**

1. Age: \_\_\_\_\_
2. Gender:  
  
☐ Male  
  
☐ Female
3. Educational Qualification:  
  

|                                     |                                           |
|-------------------------------------|-------------------------------------------|
| <input type="checkbox"/> Illiterate | <input type="checkbox"/> Secondary        |
| <input type="checkbox"/> Primary    | <input type="checkbox"/> Higher Secondary |
4. Marital Status:  
  

|                                  |                                    |
|----------------------------------|------------------------------------|
| <input type="checkbox"/> Married | <input type="checkbox"/> Unmarried |
|----------------------------------|------------------------------------|
5. Number of family members: \_\_\_\_\_
6. How many members in your family are educated? \_\_\_\_\_
7. How many members in your family are employed? \_\_\_\_\_
8. What is your role in fishing activities:

Appendix. *INTERVIEW SCHEDULE*

---

☐ Worker

☐ Shareholder

☐ Owner

9. What is your income type:

☐ Daily wages

☐ Weekly

☐ Monthly

10. What is your annual income: \_\_\_\_\_

11. What is your economic status:

☐ Lower Class

☐ Upper Middle Class

☐ Lower Middle Class

☐ High Class

☐ Middle Class

12. What is the average monthly expenditure: \_\_\_\_\_

13. My income is stable during the normal season:

☐ Strongly Agree

☐ Disagree

☐ Agree

☐ Strongly Disagree

☐ Neutral

14. My income is stable during the off-season:

☐ Strongly Agree

☐ Disagree

☐ Agree

☐ Strongly Disagree

☐ Neutral

15. Is there any savings or investments from this income:

☐ Yes

☐ No

16. For which purpose does your household spend the largest portion of its income:

- |                                             |                                            |
|---------------------------------------------|--------------------------------------------|
| <input type="checkbox"/> Food and groceries | <input type="checkbox"/> Loan repayment    |
| <input type="checkbox"/> Fishing expenses   | <input type="checkbox"/> House maintenance |
| <input type="checkbox"/> Education          | <input type="checkbox"/> Social functions  |
| <input type="checkbox"/> Health expenses    | <input type="checkbox"/> Savings           |

17. What is your mode of fishing:

- |                                      |                                     |
|--------------------------------------|-------------------------------------|
| <input type="checkbox"/> Traditional | <input type="checkbox"/> Mechanized |
|--------------------------------------|-------------------------------------|

18. Years of experience: \_\_\_\_\_

19. I have adopted new fishing techniques:

- |                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <input type="checkbox"/> Strongly Agree | <input type="checkbox"/> Disagree          |
| <input type="checkbox"/> Agree          | <input type="checkbox"/> Strongly Disagree |
| <input type="checkbox"/> Neutral        |                                            |

20. I am equipped with climate information devices:

- |                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <input type="checkbox"/> Strongly Agree | <input type="checkbox"/> Disagree          |
| <input type="checkbox"/> Agree          | <input type="checkbox"/> Strongly Disagree |
| <input type="checkbox"/> Neutral        |                                            |

21. Fishing is safer now:

- |                                         |                                   |                                            |
|-----------------------------------------|-----------------------------------|--------------------------------------------|
| <input type="checkbox"/> Strongly Agree | <input type="checkbox"/> Neutral  | <input type="checkbox"/> Strongly Disagree |
| <input type="checkbox"/> Agree          | <input type="checkbox"/> Disagree |                                            |

22. Fishermen are aware of boundaries:

- |                                         |                                   |                                            |
|-----------------------------------------|-----------------------------------|--------------------------------------------|
| <input type="checkbox"/> Strongly Agree | <input type="checkbox"/> Neutral  | <input type="checkbox"/> Strongly Disagree |
| <input type="checkbox"/> Agree          | <input type="checkbox"/> Disagree |                                            |



23. Do you receive government welfare schemes:

☐ Yes

☐ No

24. Government helping fishermen in danger:

☐ Yes

☐ No

25. Measures to improve safety:

☐ Safety helpline

☐ Safety gadgets

☐ Awareness programs